

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Engineered pluggable **distributed executor** routing pipeline nodes to **Docker** or **AWS Lambda**, backed by **S3** artifact store eliminating redundant recomputation.
- Designed stateful recovery via per-node **Redis checkpointing**; interrupted workflows **resume from the last committed step** across **distributed execution**.
- Wrote systems-level code in **Go**, **Rust**, and **C++** for **scheduling**, **resource allocation**, and **fault-tolerant placement** under GPU-scarcity constraints.
- Replaced polling with **SSE** for live ML metrics streaming, cutting redundant API calls **90%** and enabling **real-time** governance dashboards.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Built and maintained **Airflow DAGs** orchestrating multi-stage data pipelines on **GCP**, with **dependency management** and automated **failure alerting**.
- Designed **BigQuery** schemas and optimized analytical queries for ML feature pipelines, reducing compute costs through **partition pruning** and **clustering**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Designed high-throughput SQL pipelines and **distributed messaging** workflows on **AWS SQS**, cutting nightly batch aggregation from **8 minutes to 90 seconds** via indexed views and partitioning.
- Built and operated **25+ production microservices** in **C++** and SQL-backed **distributed systems** for **1M+ users**, sustaining **99.9% uptime** over 4 years on **AWS** (EC2, RDS, SQS).
- Standardized failure-handling contracts across **25+ financial services**, reducing production escalations **60%** in one quarter via consistent **retry**, **timeout**, and **propagation** semantics.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- LangGraph** agent decomposes multi-table questions into dependency-ordered **SQL subqueries** with intermediate result passing; deployable as **MCP-compatible** server.

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator modeling subnet routing and security groups.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph** multi-agent system diagnosing **production incidents** via tool-use chains; routes alerts to specialized sub-agents and synthesizes **root cause reports**, reducing **MTTR**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Rust, Go, SQL, Java, HTML/CSS

Platform: Schedulers, Execution Harness, Control Plane, GraphQL, REST APIs

Infra: Kubernetes, Docker, AWS, GCP, Kafka, PostgreSQL, Snowflake, Terraform

Practices: Distributed Systems, Data Structures and Algorithms, Design Docs, Code Review

CERTIFICATIONS

AWS Certified Developer Associate | GCP Professional MLOps Engineer